

Date: ?
14/5/26

NAME | NAND LAL

PROCEDURE FOR LIQUID PENETRANT EXAMINATION

85%
Pass.
13/5/26

- ✓ 1) **SCOPE:** This procedure covers the method for carrying out Liquid Penetrant Testing to detect surface breaking defects in non-porous materials such as metals and plastic
- ✓ 2) **PURPOSE:** The purpose of Liquid Penetrant Testing is to locate and reveal surface cracks, seams, laps, and porosity without breaking the test component.
- ✓ 3) **REFERENCES:**
 - ✓ ASME Code Section V -applicable edition/addenda.
 - ✓ ASME Code Section I -applicable edition/addenda.
 - ✓ ASME Code Section VIII, Div1-applicable edition/addenda.
 - ✓ ANSI/ASME B31.1, B31.3 applicable edition/addenda.
 - ✓ ASTM-E165-Standard Test Method for Liquid Penetrant Examination.
 - ✓ Client specification requirements for acceptance criteria.
- ✓ 4) **DEFINATIONS:**
 - Penetrant:** A liquid used to enter surface defects by Capillary Action.
 - Developer:** A material used to draw penetrant out of defects.
 - Dwell Time:** Time allowed for penetrant to penetrate defects.

✓ **5) RESPONSIBILITIES: NDT Inspector:** Perform testing as per procedure and standards

✓ **6) Equipment and Materials:** Solvent Removable Visible Dye Penetrant materials consist of:

- a) Penetrant. b) Penetrant remover / cleaner. c) Developer. d) Lint free cloth. e) The meter capable of measuring visible light intensity in lux / foot candle.

✓ **METHOD: SOLVENT REMOVABLE**

✓ **7) PROCEDURE:**

- ✓ **1) Surface Condition:** The surface should be smooth, clean, dry, and free from oil, grease, paint, rust, or dust.
- ✓ **2) Surface Cleaning:** Clean the surface properly using a cleaner and let it dry.
- ✓ **3) Apply Penetrant:** Spray the liquid penetrant on the surface so it covers the test area.
- ✓ **4) Waiting Time (Dwell Time):** Leave the penetrant on the surface for some time so it can go into surface cracks.
- ✓ **5) Remove Extra Penetrant:** Wipe off the extra penetrant from the surface carefully.
- ✓ **5) Apply Developer:** Apply developer to pull the penetrant out from cracks and make defects visible.
- ✓ **7) Development Time:** Wait 10 minutes for indications to appear.
- ✓ **8) Inspection:** Check the surface under proper light to see cracks or defects.

✓ 9) **Post Cleaning:** Clean the surface after inspection.

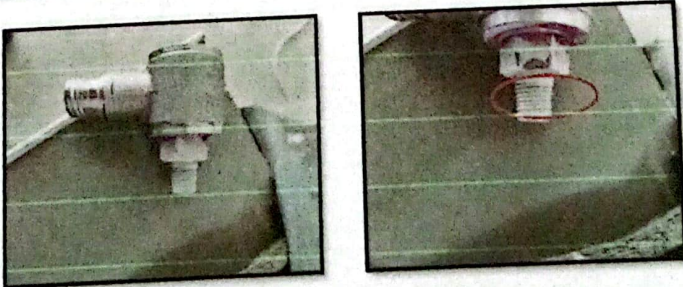
✓ **Examination and Evaluation:** No relevant indication found.

✓ **Results:** Found to be free of surface defects.

✓ **7) DOCUMENTATION:** Results are recorded in documentation sheet.

✓ **8) RECORD:** Liquid Penetrant Examination shall be documented on
Liquid Penetrant Examination Report form no: cert-PT-01

LIQUID PENETRANT INSPECTION REPORT

Customer NURICON WIRELINE SERVICES		Date of Work 14-05-2026
Rig & Well No.		W.O. No. 9493
Location NURICON YARD HYDERABAD		Cert. No. PTIS-DPT-9493-001
SGT		
Material / Item Type and Description		PO:
Material / Item serial No. 1222543		Standard / Specification ASTM E-165
Type of Inspection DYE PENETRANT TESTING		Validity of Inspection:
Date of Inspection 14-05-2026		
Surface Preparation	☐ Buffing ☐ Grinding ☒ Wire Brush ☐ Machining ☒ Wipe Clean ☐ As Blasted ☐ Removed Paint ☐ Part Dismantled ☒ Part Not Dismantled	
Technique	Type of Penetrant	Type of Developer
☒ Visible Penetrant ☐ Fluorescent Penetrant	☒ Solvent Removable Penetrants. ☐ Water-Washable Penetrants. ☐ Postemulsification Penetrants (Lipophilic) ☐ Postemulsification Penetrants (Hydrophilic)	☐ Dry Developer ☒ Wet Developer (Aqueous) ☒ Wet Developer (Nonaqueous)
Dwell Time		
Penetrant Time 10 minutes		
Developing Time 05 minutes		
Testing Medium		
☒ Wet ☐ Dry (Iron Powder) ☐ Fluorescent	Treatment Type Nabakem Penetrant Type Nabakem Developer Type Nabakem	Colour : Colourless Colour : Red Colour : White
Application : Spray		
Consumable Traceability		
Inspection Consumable	Manufacturer	Batch No.
Treatment	NABAKEM	240403/2B
Penetrant	NABAKEM	241021/7
Developer	NABAKEM	230828/7
		Expiry Date
		JAN-2027
		APR-2027
		JUN-2026
		
Area of inspection:	FULL BODY INSPECTED	
Results :	FOUND TO BE CRACKS AT THE TIME OF INSPECTION.	
Recommendation/ Comments:	NOT SATISFACTORY FOR USE.	
INSPECTED BY NAND KUMAR MOTIANI Inspector Level II MT,PT,UT		

PTIS # DPT -01/REV-03/20.05.2011



Corporate Partner

